Preview: set-01/by_level/Grade_03-04_Ecolier/y2024_q20 (v1)

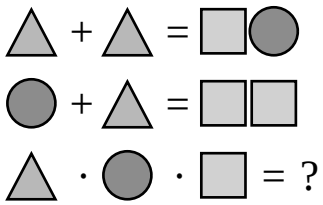
Cards hiding digits: find the forced digit first (two digits add to at most 18), then try and check

How to think about it: cards that hide digits

Each shape hides one digit (0 to 9). The same shape always hides the same digit, and different shapes hide different digits. Two cards side by side are a **two-digit number**: if the square is 1 and the circle is 4, then [square][circle] is 14.

The big idea: don't start guessing everywhere. First find the card that can only be **one** digit. Then try the few digits that are left, and **check every calculation**.

Step by step: the contest question



- **1. Find the forced card.** The biggest two digits are 9 + 9 = 18. So when two digits add up to a two-digit number, it starts with **1**. That means the square is **1**.
- **2. Write what you know.** Now the second line says circle + triangle = 11, and the first line says triangle + triangle = 1circle (a number from 10 to 18).
- **3. Try the digits that are left in a table.** Triangle + triangle must be 10 or more, so the triangle is 5, 6, 7, 8 or 9.

triangle	triangle + triangle	so circle is	circle + triangle	is it 11?
5	10	0	5	no
6	12	2	8	no
7	14	4	11	yes!
8	16	6	14	no
9	18	8	17	no

- **4. Only one row works:** triangle = 7, circle = 4, square = 1. Check: 7 + 7 = 14 and 4 + 7 = 11. Both are right.
- **5. Answer the question.** 7 · 4 · 1 = **28**. (The dot · means times.)

Watch out

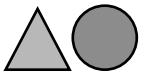
- Different shapes = different digits. If two shapes come out the same, that try is wrong.
- A two-digit number never starts with 0.
- Always check your digits in **every** calculation, not only in one.

Try it yourself: make your own puzzle. Pick digits for a triangle and a circle, write two sums with the answers, then cover the digits with paper shapes. Can someone else solve it?

Warm-up: small steps first (no points)

W1. The triangle hides the digit 9 and the circle hides the digit 8.

Which number do these two cards make?



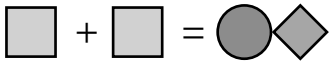
A. 17 ***B.*** 72 ***C.*** 89 ***D.*** 98 ***E.*** 908

W2. What is the **biggest** number you can get by adding two **different** digits?

A. 9 ***B.*** 13 ***C.*** 14 ***D.*** 16 ***E.*** 17

W3. Cards of the same shape hide the same digit, different shapes hide different digits.

Which digit is hidden under the **circle**?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 5 ***E.*** 9

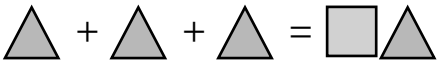
W4. The diamond hides 8. It is true that diamond + diamond = 1 followed by the triangle (a number between 10 and 19).

Which digit is hidden under the triangle?

A. 1 ***B.*** 4 ***C.*** 6 ***D.*** 7 ***E.*** 8

W5. Cards of the same shape hide the same digit, different shapes hide different digits.

Which digit is hidden under the **triangle**?

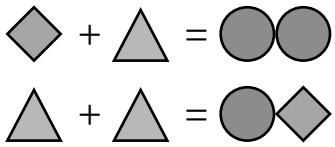


A. 0 ***B.*** 1 ***C.*** 3 ***D.*** 5 ***E.*** 9

Part A: Each question is worth 3 points

1. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Tom lays out the cards so that the calculations are correct.

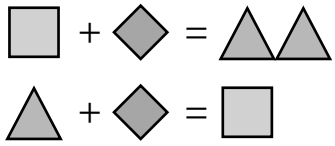
Which digit is hidden under the **diamond**?



A. 1 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

2. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Leo lays out the cards so that the calculations are correct.

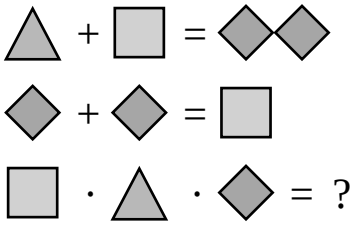
Which digit is hidden under the **diamond**?



A. 1 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 9

3. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Kim lays out the cards so that the calculations are correct.

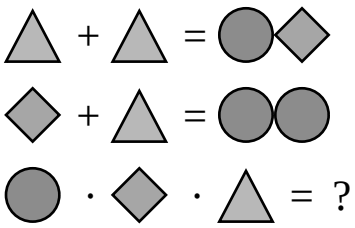
What number does Kim get for the last calculation?



A. 0 ***B.*** 18 ***C.*** 19 ***D.*** 22 ***E.*** 36

4. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Tom lays out the cards so that the calculations are correct.

What number does Tom get for the last calculation?

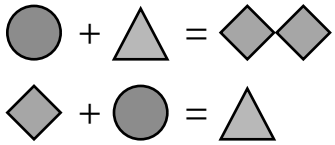


A. 0 ***B.*** 12 ***C.*** 27 ***D.*** 28 ***E.*** 29

Part B: Each question is worth 4 points

5. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Ben lays out the cards so that the calculations are correct.

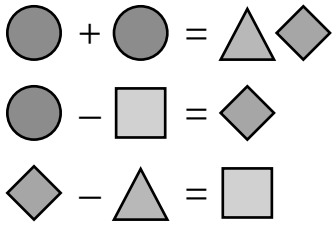
Which two-digit number do the cards diamond and triangle make?



A. 16 ***B.*** 17 ***C.*** 20 ***D.*** 26 ***E.*** 61

6. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Ben lays out the cards so that the calculations are correct.

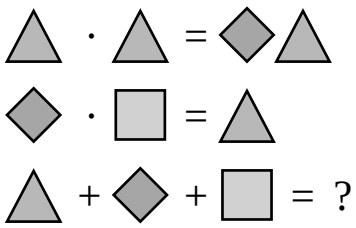
Which two-digit number do the cards square and circle make?



A. 36 ***B.*** 37 ***C.*** 38 ***D.*** 47 ***E.*** 73

7. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Tom lays out the cards so that the calculations are correct.

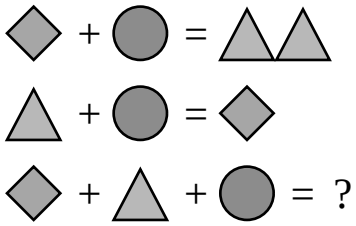
What do you get if you add the digits of all three shapes?



A. 10 ***B.*** 11 ***C.*** 12 ***D.*** 13 ***E.*** 14

8. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Mia lays out the cards so that the calculations are correct.

What do you get if you add the digits of all three shapes?

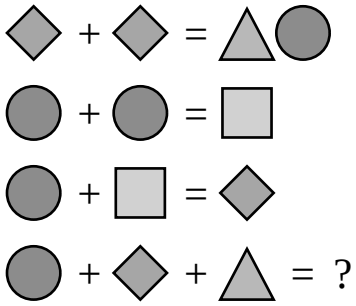


A. 9 ***B.*** 11 ***C.*** 12 ***D.*** 13 ***E.*** 14

Part C: Each question is worth 5 points

9. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Leo lays out the cards so that the calculations are correct.

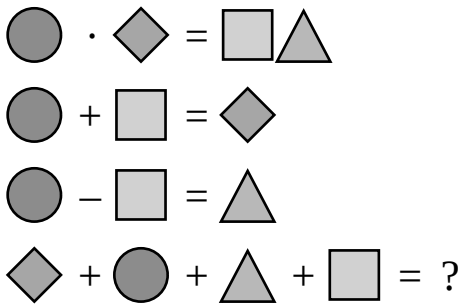
What number does Leo get for the last calculation?



A. 8 ***B.*** 9 ***C.*** 10 ***D.*** 11 ***E.*** 18

10. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Lucas lays out the cards so that the calculations are correct.

What number does Lucas get for the last calculation?



A. 8 ***B.*** 10 ***C.*** 11 ***D.*** 12 ***E.*** 19

Answer Key

W1.	**D**	**W2.**	**E**	**W3.**	**B**	**W4.**	**C**	**W5.**	**D**	**1.**	**C**
2.	**C**	**3.**	**B**	**4.**	**D**	**5.**	**A**	**6.**	**B**	**7.**	**B**
8.	**C**	**9.**	**B**	**10.**	**B**						

Solutions

W1. D (98) — The left card is the tens digit, the right card the ones digit: 9 tens and 8 ones = 98. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · w_read_number]

W2. E (17) — 9 + 8 = 17 (9 + 9 is not allowed, the digits must be different). [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · w_biggest_sum]

W3. B (1) — Square + square is at most 9 + 9 = 18. A two-digit answer from 10 to 18 starts with 1, so the circle is 1. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · w_forced_one]

W4. C (6) — 8 + 8 = 16, so the triangle is 6. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · w_one_step]

W5. D (5) — Try the digits: 3 × triangle must end with the same digit as the triangle. 5 + 5 + 5 = 15 works (and 0 does not, because then the square would be 0 too). So it is 5. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · w_three_same]

1. C (4) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Then try the possible digits and check both calculations: triangle = 7, circle = 1, diamond = 4. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · find_digit]

2. C (5) — Two digits add up to at most 9 + 9 = 18, so the triangle at the front of the two-digit answer must be 1. Then try the possible digits and check both calculations: diamond = 5, square = 6, triangle = 1. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · find_digit]

3. B (18) — Two digits add up to at most 9 + 9 = 18, so the diamond at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: triangle = 9, square = 2, diamond = 1. So 2 · 9 · 1 = 18. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · product_all]

4. D (28) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: circle = 1, diamond = 4, triangle = 7. So 1 · 4 · 7 = 28. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · product_all]

5. A (16) — Two digits add up to at most 9 + 9 = 18, so the diamond at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: circle = 5, diamond = 1, triangle = 6. So the diamond is the tens digit and the triangle the ones digit: 16. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · plus_minus]

6. B (37) — Two digits add up to at most 9 + 9 = 18, so the triangle at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: triangle = 1, diamond = 4, square = 3, circle = 7. So the square is the tens digit and the circle the ones digit: 37. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · plus_minus]

7. B (11) — Try the possible digits (start with the multiplication: which times-table result has these digits?) and check every calculation: triangle = 6, diamond = 3, square = 2. 6 + 3 + 2 = 11. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · times]

8. C (12) — Try the possible digits (start with the multiplication: which times-table result has these digits?) and check every calculation: diamond = 6, triangle = 1, circle = 5. 6 + 1 + 5 = 12. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · times]

9. B (9) — Two digits add up to at most 9 + 9 = 18, so the triangle at the front of the two-digit answer must be 1. Start with the calculation where the fewest digits are possible, then check all three: circle = 2, square = 4, triangle = 1, diamond = 6. 2 + 6 + 1 = 9. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · four_cards]

10. B (10) — Start with the calculation where the fewest digits are possible, then check all three: triangle = 2, diamond = 4, circle = 3, square = 1. 4 + 3 + 2 + 1 = 10. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · four_cards]